

Business Problem Document

Netflix Data Cleaning & Analysis

1. Background

Netflix's content catalog data, as exported for public analysis, stores multiple values inside single columns — a title's genres, countries of production, and directors are often listed together in one comma-separated text field. This structure works for casual browsing but makes real analysis difficult: it's not possible to reliably count titles per genre, join on director, or filter by country using standard SQL grouping and joins until the data is restructured. The raw data also contains missing values and duplicate entries that would distort any analysis performed on top of it.

2. Business Problem

Before any content-strategy questions can be answered — which genres dominate the catalog, which countries produce the most content, which directors are most prolific, how content acquisition has trended over time — the underlying data needs to be cleaned, deduplicated, and restructured into a proper relational format. This project addresses that gap: transforming a single messy CSV export into a clean, queryable database structure, then using it to answer real content and catalog questions.

3. Objectives

- Clean the raw dataset: remove duplicate titles and handle missing values in director, country, cast, and duration fields
- Split multi-value columns (genre, country, director) into separate normalized tables to enable proper relational querying
- Load the cleaned, normalized tables into a MySQL database
- Answer 10 business questions using SQL, ranging from basic aggregates to window functions
- Build an interactive Power BI dashboard visualizing catalog composition and trends

4. Data Source

Kaggle "Netflix Movies and TV Shows" dataset (approximately 8,800 titles), covering type, title, director, cast, country, date added, release year, content rating, duration, and listed genres. This is a publicly available dataset used for portfolio and educational purposes.

5. Scope

This project covers data cleaning, deduplication, and normalization (Python), relational business-question analysis (SQL), and dashboard visualization (Power BI). It does not include predictive modeling, sentiment analysis of descriptions, or real-time catalog updates.